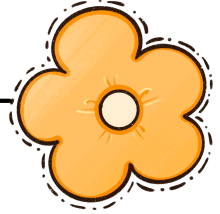


HASHMAP 2



"Opportunities don't happen, you create them."

~ Chris Grosser



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Good

Evening ☺

Today's content

- length of longest subarr with sum = 0
- Equal no. of 1's & 0's in a subarr
- Count pairs
- Count of subarr with sum = 0 (last class HW)

01 Given $arr[N]$ elements, return length of longest subarray with

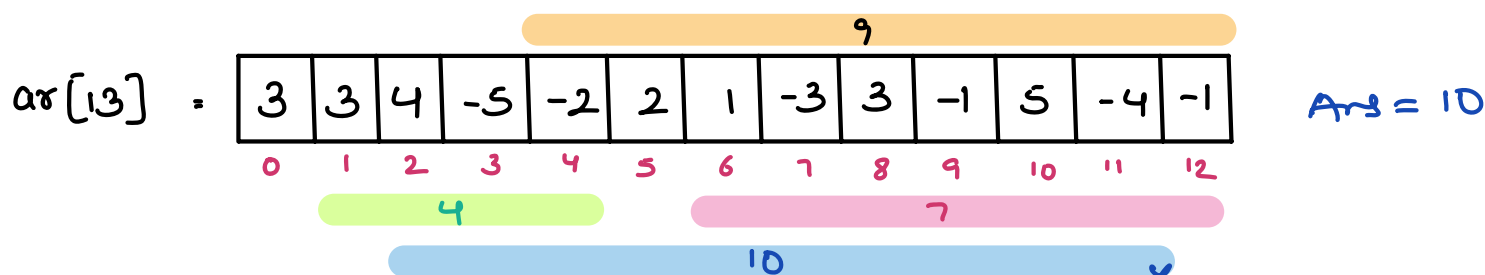
sum = 0

Constraints

$$1 \leq N \leq 10^5$$

$$1 \leq arr[i] \leq 10^9$$

Ex:-



Idea $\rightarrow$ Generate all the subarrays & check if sum == 0, compare the length with the max & update accordingly

$$TC = O(n^3) \longrightarrow O(n^2) \quad SC: O(1)$$

$$\text{Max } n = 10^5 \longrightarrow (10^5)^2 \rightarrow 10^{10} \quad \text{TLE}$$

Idea 2 :-

$arr[13] =$

3	3	4	-5	-2	2	1	-3	3	-1	5	-4	-1
0	1	2	3	4	5	6	7	8	9	10	11	12

$psum =$

3	6	10	5	3	5	6	3	6	5	10	6	5
0	1	2	3	4	5	6	7	8	9	10	11	12

Obs 1 $\rightarrow$ To find the length of the subarr

Task 2 $\rightarrow$ To make the length as max as possible

Eg1 $\rightarrow$ $pf[4] = 3$ & $pf[7] = 3$ $\#$ $7 - 4 = 3$

$$pf[7] = 3$$

$$sum[0 \ 7] = sum[0 \ 4] + sum[5 \ 7]$$

$$\cancel{3} = \cancel{3} + sum[5 \ 7]$$

$$sum[5 \ 7] = 0$$

$$subarray[5 \dots 7] = \# \text{ len} = 3$$

$$a = 5$$

$$b = 7$$

$$len = b - a + 1$$

$$= 7 - 5 + 1 = \hat{3}$$

eg 2

$$pf[5] = 5$$

$$pf[9] = 5$$

$$\#len = 4$$

$$pf[9] = sum[0 \ 9]$$

$$sum[0 \ 9] = sum[0 \ 5] + sum[6 \ 9]$$

$$5 = 5 + sum[6 \ 9]$$

$$5 - 5 = sum[6 \ 9]$$

$$sum[6 \ 9] = 0$$

$$\downarrow$$
$$subarr[6 \ 9] \rightarrow \#len = 4$$

$$* \quad pf[i] = x \quad \& \quad pf[j] = x$$

$$\#len = j - i$$

$$pf[j] = sum[0 \ j]$$

$$sum[0 \ j] = sum[0 \ i] + sum[i+1 \dots j]$$

$$x = x + sum[i+1 \dots j]$$

$$sum[i+1 \dots j] = 0$$

$\downarrow$

$$subarr[i+1 \dots j] = j - (i+1) + 1$$

$$\Rightarrow j - i - 1 + 1 = j - i$$

Obs 2 $\rightarrow$ For every ele, you need the first occurrence(index) to get max length

Store $\rightarrow$ HM { key. value }

$\swarrow$
psum

$\searrow$
first index of
psum



psum =

3	6	10	5	3	5	6	3	6	5	10	6	5
0	1	2	3	4	5	6	7	8	9	10	11	12

ans = 0 HM = { <3,0> <6,1>, <10,2> <5,3> }

<u>ele</u>	<u>curr_idx</u>	1 st occurrence of ele in HM	len = curr_idx - 1 st occur	ans
3	0	3 is not present, insert <3,0> in hm		
6	1	6 is not present, insert <6,1> in hm		
10	2	10 is not present, insert <10,2> in hm		
5	3	5 is not present, insert <5,3> in hm		
3	4	3 is present, 0	len = 4 - 0 = 4	ans = 4
5	5	5 is present, 3	len = 5 - 3 = 2	ans = 4

6 6 6 is present, | len = 6 - 1 ans = S
 = 5

3 7 3 is present, 0 len = 7 - 0 ans = 7
 = 7

9

9

כ

9

{ Dry run the rest of array }

Edge case

arr[] = { 4 -3 -1 2 -2 }

psum =

4	1	0	2	0
0	1	2	3	4

hm = { <0, -1> <4, 0> <1, 1> }

Way 1 → if psum == 0 → len = curridx + 1

Way 2 → initially in hm, we can put (0, -1)

```
int lengthSub (int []arr)
```

```
long sum = 0 // cf instead of creating psum array
```

```
HashMap < long, Integer > hm = new HashMap <> ();
```

```
ans = 0;
```

```
hm.put (0, -1);
```

TC = O(n)

```
for (i = 0; i < n; i++)
```

SC = O(n)

```
    sum = sum + arr[i];
```

```
    if (hm.containsKey(sum)) {
```

```
        int first = hm.get(sum);
```

```
        len = i - first;
```

```

    ans = Math.max(ans, len);
  }
  else {
    hm.put(sum, i);
  }
}
return ans;
}

```

Q2. Given $arr[N]$ containing only 0's & 1's. Find **max length of subarray** which contains **equal 0's & 1's**

Constraints $1 \leq N \leq 10^5$

Ex 1 :-

0	1	2	3	4	5
0	0	0	1	1	0

 $len = 4$

Ex 2 :-

0	1	2	3	4	5	6	7	8
0	1	0	0	0	1	1	0	1

 $len = 8$

Ex 3 :-

0	1	2	3	4	5	6	7	8	9	10	11
0	1	0	0	0	1	0	0	1	1	0	0

 $len = 6$

6

Idea → In given arr of 1's & 0's, replace all the 0's with -1's

⇒ max len subarray of equal no. of 1's & -1's

Ex 2 :-

0	1	2	3	4	5	6	7	8
0	1	0	0	0	1	1	0	1

0 → -1's :-

-1	1	-1	-1	-1	1	1	-1	1
----	---	----	----	----	---	---	----	---

psum ⇒

-1	0	-1	-2	-3	-2	-1	-2	-1
----	---	----	----	----	----	----	----	----

HM = { $\underbrace{\langle 0, -1 \rangle}_{\text{base case}}$ $\langle -1, 0 \rangle$ $\langle -2, 3 \rangle$ $\langle -3, 4 \rangle$ }

{ TODO }

10:24 → 10:32 pm

{
→ Interviewbit
→ leet code
→ Gfg
}

03. Given an array of N elements & K , calculate no. of pairs (i, j) such that $arr[i] + arr[j] = K$ & $(i \neq j)$

Note :- i & j are indices

Constraints $\rightarrow 1 \leq N \leq 10^5$
 $1 \leq arr[i] \leq 10^9$

Ex 1 :- $arr[] =$

0	1	2	3	4
8	5	2	5	8

$K=10$

i	j	$arr[i] + arr[j] = 10$
0	2	$8 + 2 = 10$
1	3	$5 + 5 = 10$
2	4	$2 + 8 = 10$

Ans = 3

Ex 2 :- $arr[] =$

0	1	2	3	4
4	6	3	4	7

$K=10$

i	j	$arr[i] + arr[j]$
0	1	$4 + 6 = 10$
1	3	$6 + 4 = 10$
2	4	$3 + 7 = 10$

} Ans = 3

Idea → Check for all pairs but do keep
in mind that $i \neq j$

```
for (i=0; i<n; i++)
```

```
    → for (j=0; j<i; j++) {
```

```
        if (ar[i] + ar[j] == k) { c = c+1; }
```

```
    }
```

```
}
```

TC = $O(n^2)$

SC = $O(1)$

same code

```
for (i=0; i<n; i++)
```

```
    a = ar[i] , b = k - a;
```

```
    for (j=0; j<i; j++) {
```

```
        if (ar[j] == b) { c = c+1; }
```

```
    }
```

```
}
```

count
the b
in [0 i-1]

$$arr[7] =$$

0	1	2	3	4	5	6
4	6	<u>2</u>	4	6	2	8

$$K=10$$

i	(a) arr[i]	(b) K-a	# count of b in [0 i-1]
0	4	6	# count of 6 [0 -1] = 0
1	6	4	# count of 4 [0 0] = 1
2	2	8	# count of 8 [0 1] = 0
3	4	6	# count of 6 [0 2] = 1
4	6	4	# count of 4 [0 3] = 2
5	2	8	# count of 8 [0 4] = 0
6	8	2	# count of 2 [0 5] = 2

Ans = 6

Final Idea $\rightarrow$ Optimise the searching & counting of b, we need HashMap {key, value}
 $\downarrow$
 element $\rightarrow$ count

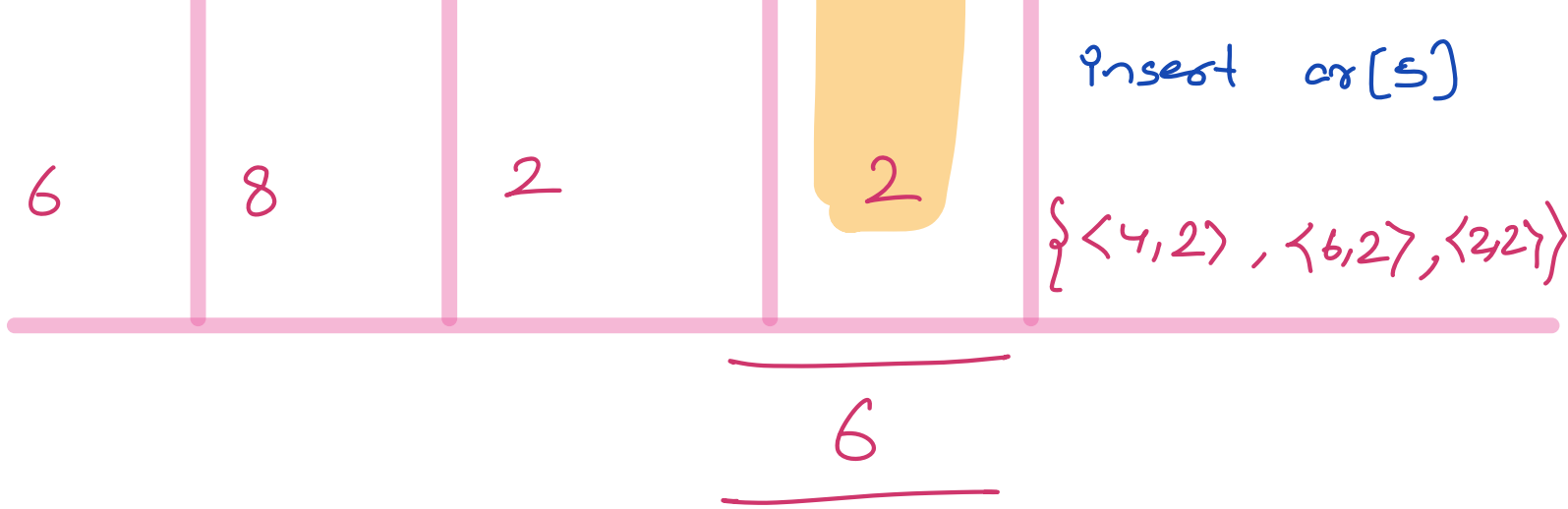
For an index i , we need the values in hm from $[0 \text{ to } i-1]$

$$ar[7] =$$

0	1	2	3	4	5	6
4	6	<u>2</u>	4	6	2	8

$$K=10$$

i	$ar[i]$	$K - ar[i]$	# count of b in HM	HM
0	4	6	0	{ } insert $ar[0]$
1	6	4	1	{ <4,1> } insert $ar[1]$
2	2	8	0	{ <4,1>, <6,1> } insert $ar[2]$
3	4	6	1	{ <4,1>, <6,1>, <2,1> } insert $ar[3]$
4	6	4	2	{ <4,2>, <6,1>, <2,1> } insert $ar[4]$
5	2	8	0	{ <4,2>, <6,2>, <2,1> }



Steps

01. figure out a & b
02. look for the count of b in HM
03. insert the ele vs count in hashmap
for $\{0 \dots i-1\}$

```
int countPairs (int [] arr, int k)
```

```
HashMap < Integer, Integer > hm = new HashMap<>();
```

arr[i] count

```
int ans = 0
```

```
for (i = 0; i < n; i++)
```

```
    a = arr[i]
```

```
    b = k - arr[i]
```

```
    if (hm.containsKey(b)) {
```

```
        |   ans += hm.get(b);
```

```
        |
```

```
        // insert arr[i] in hm
```

```
    if (hm.containsKey(a)) {
```

```
        |   int f = hm.get(a);
```

```
        |   hm.put(a, f + 1);
```

```
        |
```

```
    else {
```

```
        |   hm.put(a, 1);
```

```
        |
```

```
    }
```

```
return ans;
```

} updating
the ans

} Modify the
hm for
next iteration

TODO $\rightarrow \{ \text{arr}[i] - \text{arr}[j] == k \} \ \& \ (i \neq j)$

count the pairs

Q Find the count of subarray in the array which sums up to zero

$\left\{ \begin{array}{l} \text{arr}[5] : \begin{array}{c|cccc} 0 & 1 & 2 & 3 \\ \hline 2 & 4 & -2 & -2 \end{array} \\ \text{psum} : \begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 2 & 6 & 4 & 2 \end{array} \end{array} \right\} \text{count} = 1$

$\left\{ \begin{array}{l} \text{arr}[5] : \begin{array}{c|cccccc} 0 & 1 & 2 & 3 & 4 & 5 \\ \hline 2 & 4 & -2 & -2 & 3 & -3 \end{array} \\ \text{psum} : \begin{array}{c|cccccc} & 0 & 1 & 2 & 3 & 4 & 5 \\ \hline 2 & 6 & 4 & 2 & 5 & 2 \end{array} \end{array} \right\} \text{count} = 3$

$\left\{ \begin{array}{l} \text{arr}[4] : \begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 1 & -1 & -2 & 2 \end{array} \\ \text{psum} : \begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 1 & 0 & -2 & 0 \end{array} \end{array} \right\} \text{count} = 3$

HM $\rightarrow \{ \text{psum}, \text{count} \}$

$\text{HM} = \left\{ \begin{array}{l} \langle 0, 3 \rangle \\ \langle 1, 1 \rangle \\ \langle -2, 1 \rangle \end{array} \right\}$

 (Edge case)

ans = 0 + 1 = 1

HM

→ Edge case `hm.put(0, 1)`

→ If `psum` is not present in `hm`

↳ update the `hm.put(psum, 1);`

→ If `psum` is present

`ans += hm.get(psum);`

update the freq of `psum` in `hm`

↳ `hm.put(psum, of + 1);`

Doubt session

$$\text{sum} = k$$

$$\text{sum} = 0$$

if a sum is repeating then
we have a subarr with $\text{sum} = 0$

$$x + y = k$$

$$k = 5$$

1 2 3 4 5

$$\begin{array}{cccccc} \text{psum} = & \underline{1} & 3 & 6 & 10 & 15 \\ & 0 & 1 & 2 & 3 & 4 \end{array}$$

$$\langle \underline{1}, \underline{0} \rangle \langle 3, 1 \rangle \langle 6, 2 \rangle$$
$$\langle \underline{10}, \underline{3} \rangle$$

$$\text{psum} - k$$

$$\text{sum} = 5$$

$$1 - 5 = \underline{-4} \quad \text{psum} = \underline{1}$$

$$3 - 5 = \underline{-2} \quad \text{subarr sum} = -2$$

$$6 - 5 = \underline{1} \quad \text{subarr sum} = 1 \rightarrow 0^+ \text{ index}$$

$$\underline{\underline{\text{len} = 2 - 0 = 2}}$$

$$10 - 5 = 5$$

$$15 - 5 = \underline{10} \rightarrow \text{Do we have subarr sum up to be 5}$$

$$\text{len} = 4 - 3 \Rightarrow 1$$

$$\text{arr} = \{ 1, 2, 2, 3, 3 \}$$

$$\boxed{B = 2}$$

$$\text{HM} = \left\{ \begin{array}{l} 1 \rightarrow 1 \\ 2 \rightarrow 2 \\ 3 \rightarrow 2 \end{array} \right\}$$

$$\text{Ans} = 5$$

* Count unique ele

$$A = \{ 3, 4, 3, 6, 6 \}$$

$$B = \{ 3, 3, 3, 9, 0, 1, 0 \}$$

$\text{map.put}(\underline{\text{key}}, \underline{\text{value}}) \rightarrow$ insert if key is not present
if it is present, update val

$\text{map.get}(\text{key}) \rightarrow$ if key is present $\rightarrow$ value
if key is not present $\rightarrow$ null

map.containsKey(key) → Return true if key is present
false if key is not present

$$A = \{ 3, 4, 3, 6, 6 \}$$

$$B = \{ \underline{3}, 3, 3, 9, 0, 4, 0 \}$$

↑

$$HM = \left\{ \begin{array}{l} \underline{3 \rightarrow 2} \\ 4 \rightarrow 1 \\ 6 \rightarrow 2 \end{array} \right\}$$

```
if (hm.containsKey(B[i])) {  
    count = count + 1  
    hm.remove(B[i]);  
}
```

3